

Chapter 2 Research Plan

Conditional Sequential Mixture of Flows (CSMF) for Imaging Inverse Problems

Plan start: 01 Oct 2025 **First outcomes draft:** 31 Mar 2026

Goal

Build a **Conditional Sequential Mixture of Flows (CSMF)** for imaging inverse problems, trained with a **hybrid objective** that combines: (i) likelihood/NLL, (ii) measurement consistency with the forward model, (iii) a geometry term (e.g., sliced-Wasserstein SW_2 or MMD), and (iv) a light calibration term (e.g., CRPS or temperature scaling). Demonstrate end-to-end gains on **optical super-resolution (SR)** and **SAR despeckling** while preserving or improving NLL.

Why Prototype on MNIST First?

- Fast iterations on 28×28 grayscale for debugging conditional flows, gating, and consistency steps.
- Controlled inverse tasks (blur + downsample, masking) to validate that hybrid losses improve measurement consistency and geometry before moving to higher-resolution remote-sensing data.
- Enables ablations and tuning (temperature τ for gating, top- k , number of data-consistency steps) with minimal compute.

Work Packages (WPs) & Milestones

WP0 — Scope, infrastructure, baselines (01–15 Oct 2025). Refactor to support **conditional experts** (FiLM/affine conditioning by y) and a **data-dependent gate** $w_\psi(y)$. Keep three experts (RealNVP, MAF, RBIG) and start with **global gating**. Baselines: (a) single conditional flow, (b) CSMF without the hybrid loss.

Deliverable D0 (15 Oct): runnable CSMF skeleton on MNIST with unit tests (invertibility, log-det).

WP1 — Measurement-consistency layer (16 Oct–15 Nov 2025). Implement a proximal/data-consistency step between flow blocks. For SR: closed-form proximal update in Fourier for blur+downsample; for SAR: log/Anscombe variance-stabilizing transform and quadratic residual in the stabilized domain. Enable plug-and-play sampling (few alternations of flow and consistency).

Milestone M1 (15 Nov): MNIST deblurring/SR shows lower residual $\|Ax - y\|$ at matched NLL.

WP2 — Hybrid objective & training schedule (16 Nov–15 Dec 2025). Train with

$$\mathcal{L} = \text{NLL} + \lambda_{\text{cons}} \mathbb{E}[d(Ax, y)] + \lambda_{\text{trans}} (SW_2 \text{ or MMD}) + \lambda_{\text{cal}} \text{CRPS}.$$

Start $\lambda_{\text{cons}} = 0.1$, $\lambda_{\text{trans}} = 0.05$; use annealing or a two-phase schedule (NLL $\rightarrow$ hybrid fine-tune). Freeze experts and train the gate first; then a light joint fine-tune.

Milestone M2 (15 Dec): Hybrid beats pure NLL on consistency & transport without hurting NLL (MNIST).

WP3 — Ablations on MNIST (16 Dec 2025–20 Jan 2026). Gating: global vs. local (per patch), top- k vs. soft, τ annealing. Consistency: number of steps/projections, proximal weight λ . Transport: SW_2 vs. MMD; calibration: temperature scaling vs. none.
Deliverable D3 (20 Jan): ablation tables & plots; select defaults.

WP4 — Port to imaging tasks (21 Jan–10 Mar 2026). **Optical SR:** DIV2K/BSD68 grayscale patches $\rightarrow$ blur+downsample ($\times 2$, $\times 4$), AWGN; real LR via downscaling HR.
SAR despeckling: simulated Gamma speckle or multi-look references; log-domain consistency.
Metrics: NLL (when feasible), $\|Ax - y\|$, PSNR/SSIM, SAM, ERGAS, ENL, radiometric bias, SW_2 /MMD, CRPS, runtime.
Milestone M4 (10 Mar): end-to-end SR + proto-SAR matching MNIST gains.

WP5 — First outcomes & write-up (11–31 Mar 2026). Freeze seeds/configs; run final sweeps for chosen settings. Draft Chapter-2 results + short tech report/preprint (method + MNIST + SR + early SAR). Plot bank: consistency curves, SW_2 vs. NLL Pareto, calibration (PIT/CRPS), qualitative panels.
Deliverable D5 (31 Mar): first outcomes draft + reproducible repo branch.

Experiment Matrix

Task	Forward model A	Noise	Datasets	Key metrics
MNIST Deblur/SR	Gaussian blur + downsample ($\times 2$, $\times 4$)	AWGN	MNIST	NLL, $\ Ax - y\ $, SW_2 (512 proj), MMD (multi- σ), CRPS
Optical SR	Blur + downsample (kernel mismatch study)	AWGN	BSD68 / DIV2K (gray)	PSNR/SSIM, SAM, ERGAS, $\ Ax - y\ $, SW_2 /MMD, CRPS, runtime
SAR despeckling (proto)	Identity in log-domain	Gamma speckle $\rightarrow$ log	Simulated + multi-look refs	ENL, edge preser- vation, radiometric bias, residual, CRPS

Implementation Checklist (Minimal Code Changes)

- **Conditional experts:** add FiLM/affine conditioning features from a lightweight encoder $c_\phi(y)$.
- **Gating $w_\psi(y)$:** global softmax with temperature $\tau \approx 1.1$ and top- k masking; later, optional local (per-patch) gating.
- **Consistency layer:** closed-form proximal step for SR; log-domain quadratic residual for SAR.
- **Sampler:** single-shot mixture + optional few rounds of (flow $\leftrightarrow$ consistency) refinement.
- **Hybrid loss:** modular toggles for λ 's; use sliced-Wasserstein (differentiable) for geometry.
- **Logging & eval:** save seeds/configs; shared utilities for SW_2 , MMD, CRPS, residual; calibration plots (PIT).

Definition of Done (by 31 Mar 2026)

- CSMF with conditional experts + data-dependent gating + consistency layer implemented and tested.
- Hybrid objective validated: consistency & transport improved at matched/better NLL on MNIST and SR; initial SAR gains.
- Ablations completed with clear default configuration.
- Draft text + figures ready (Chapter-2 core results + preprint seed).
- Reproducible one-command script to regenerate the main tables and plots.

Risks & Mitigations

- **Gate collapse / expert underuse:** top- k gating, τ annealing, light entropy on the gate; monitor effective number of experts N_{eff} .
- **Consistency hurts NLL:** two-phase training or Lagrangian multipliers; few proximal steps.
- **Transport cost:** use sliced-Wasserstein (≤ 512 projections), avoid full OT.
- **SAR realism:** start in log-domain with simulated speckle; progress to multi-look references.

Writing & Venues

- By 31 Mar 2026: internal report / first-draft manuscript with MNIST + SR + proto-SAR.
- April–May 2026: extend SAR experiments; target IEEE TCI/TGRS journal or TMLR (rolling), with code release.

Note. Chapter 1 (AMF-VI mixtures) is assumed finalized and submitted. Chapter 2 builds directly on it with conditional, physics-aware gating and a hybrid objective.